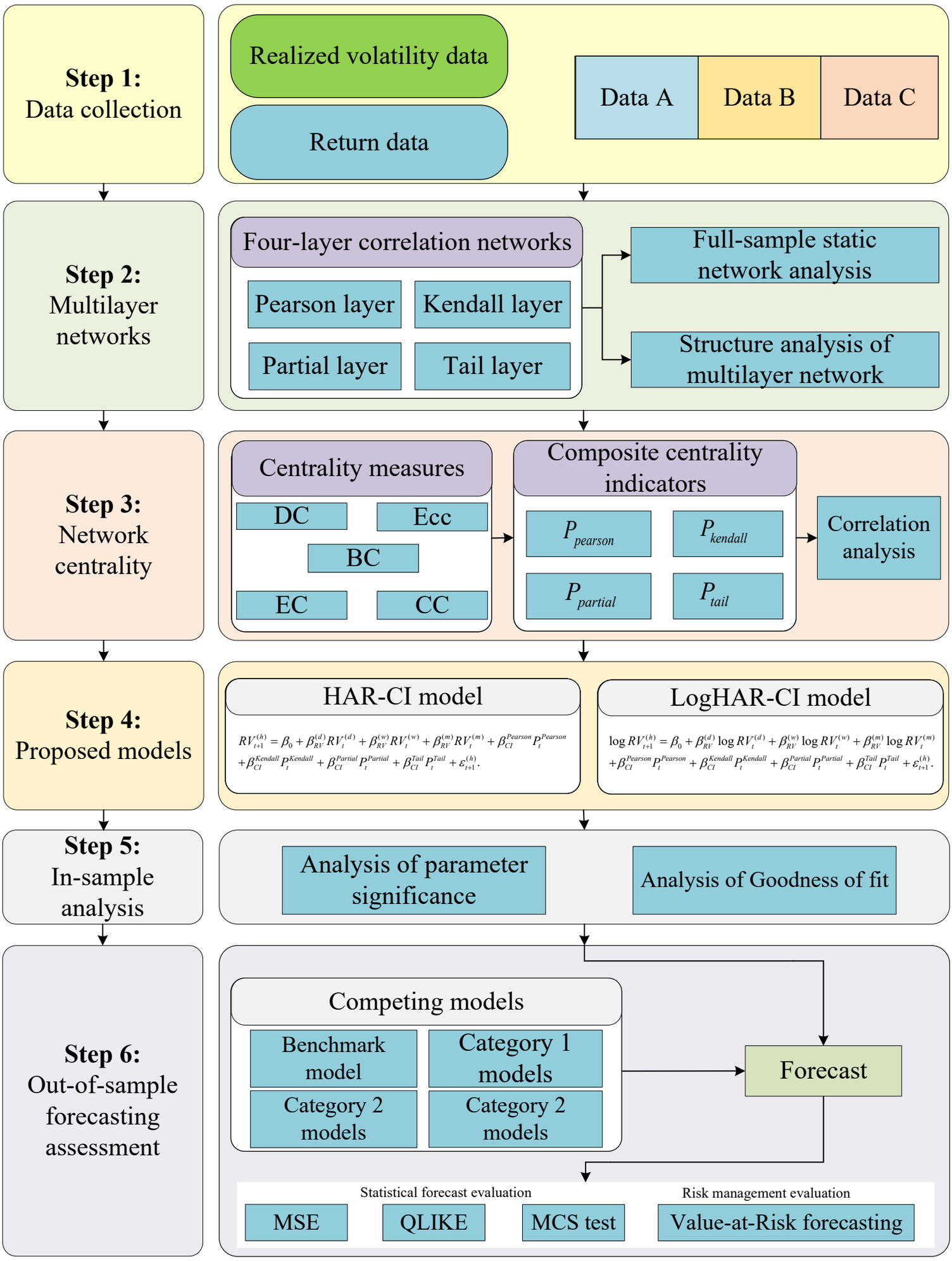

Step 1:
Data collection
Realized volatility data
Return data
Data A
Data B
Data C
Step 2:
Multilayer networks
Four-layer correlation networks
Pearson layer
Kendall layer
Partial layer
Tail layer
Full-sample static network analysis
Structure analysis of multilayer network
Step 3:
Network centrality
Centrality measures
DC
Ecc
BC
EC
CC
Composite centrality indicators
$P_{pearson}$
$P_{kendall}$
$P_{partial}$
P_{tail}
Correlation analysis
Step 4:
Proposed models
HAR-CI model
$RV_{t+1}^{(h)} = \beta_0 + \beta_{RV}^{(d)} RV_t^{(d)} + \beta_{RV}^{(w)} RV_t^{(w)} + \beta_{RV}^{(m)} RV_t^{(m)} + \beta_{CI}^{Pearson} P_t^{Pearson} + \beta_{CI}^{Kendall} P_t^{Kendall} + \beta_{CI}^{Partial} P_t^{Partial} + \beta_{CI}^{Tail} P_t^{Tail} + \varepsilon_{t+1}^{(h)}.$
LogHAR-CI model
$\log RV_{t+1}^{(h)} = \beta_0 + \beta_{RV}^{(d)} \log RV_t^{(d)} + \beta_{RV}^{(w)} \log RV_t^{(w)} + \beta_{RV}^{(m)} \log RV_t^{(m)} + \beta_{CI}^{Pearson} P_t^{Pearson} + \beta_{CI}^{Kendall} P_t^{Kendall} + \beta_{CI}^{Partial} P_t^{Partial} + \beta_{CI}^{Tail} P_t^{Tail} + \varepsilon_{t+1}^{(h)}.$
Step 5:
In-sample analysis
Analysis of parameter significance
Analysis of Goodness of fit
Step 6:
Out-of-sample forecasting assessment
Competing models
Benchmark model
Category 1 models
Category 2 models
Category 2 models
Forecast
Statistical forecast evaluation
Risk management evaluation
MSE
QLIKE
MCS test
Value-at-Risk forecasting